

MIR HASSAN

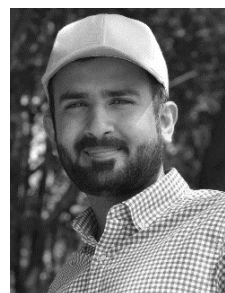
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Research Interest:

Blockchain Technology, Federated learning, Distributed Systems, and Internet of Things (IoT)

Education

- 2022 – 2025 **PhD, University of Trento** in Information Engineering and Computer Science
Thesis title: *Communication-Efficient Federated Learning for Resource-Constrained Edge Environments*
- 2017 – 2019 **Master of Engineering, Wuhan University** in Software Engineering
Thesis title: *Education Platform Based on Blockchain Technology for Verification and Validation of Certification*
- 2011 – 2014 **BS, Telecommunication, University of Sindh**
Thesis title: *Optimization of satellite link to earth station using satellite tool kit (STK)*
Collaboration: *Pakistan Space and Upper Atmosphere Research Commission (SUPARCO)*

Academic Experience

- April 2026 – Present **Research Collaborator** Department of Information Engineering and Computer Science, University of Trento, Trento, Italy
- Jan 2026 – Present **Senior Researcher/Associate Professor** Human Environment Technology (HET), Mykolas Romeris University (MRU), Vilnius, Lithuania
- April 2024 – Sept 2024 **Visiting Researcher** Department of Electronics Engineering, Yildiz Technical University, Istanbul, Turkey
- Oct 2023 – oct 2023 **Visiting Researcher** Department of Electronics Engineering, Yildiz Technical University, Istanbul, Turkey
- May 2022 – Aug 2023 **Specialist**, Department of Bio-mechanical Engineering, Vilnius Tech University, Vilnius, Lithuania **Project end: 31-Aug-2023 (Part-Time)**
- Jan 2020 – Mar 2020 **Visiting Researcher** Essence Data Engineering & Distributed Computing Systems at School of Computing Science, University of Glasgow, Scotland, United Kingdom **Remotely**

Co-Supervised

- 2024 **Hafiz Muhammad Bilal**, thesis title: *Enhancing Communication-Efficient Federated Learning for Human Activity Recognition through Highly Compressed Models*, Master's Degree in Data Science at University of Trento, Trento, Italy.

Projects

- Jan 2023 – March 2026  **HORIZON EIC PATH FINDER PROJECT, SUSTAIN (Smart Building Sensitive to Daily Sentiment, Grant agreement No 101071179), WP3, University of Trento, Trento, Italy.**
- Jan 2019 - June 2019  **Educational platform based on blockchain technology for verification and validation of certificates, Wuhan University, Wuhan, China.**
- Jan 2014 - Dec 2014  **Optimization fo satellite link to earth station using satellite tool kit (STK)**
Project Collaboration: *Pakistan Space and Upper Atmospheric Research Commission(SUPARCO)*

Research Publications

Journal Articles

- 1 **Hassan, Mir**, Nawaz, A., Ali Hassan, S., Johnsson, M., & Tafazolli, R. (2026). Semantic-aware sparse communication for federated learning in mission-critical 6g. *IEEE Open Journal of the Communications Society*. Submitted.
- 2 Ali, N., **Hassan, Mir**, Sodhro, A. H., Aloï, G., Gravina, R., Iacca, G., & Rango, F. D. (2025). Proximity-aware federated learning for symbiotic task offloading in vehicular edge intelligence. *IEEE Internet of Things Journal*, 1–1. **JUFO Level: 2.**  doi:10.1109/JIOT.2025.3591253
- 3 **Hassan, Mir**, Ali, N., Aloï, G., Gravina, R., Fortino, G., Yildirim, K. S., & Iacca, G. (2025). Tinyml at the wireless edge: Architectures, deployment modes, and 6g integration. *IEEE Wireless Communications*. Under Review. **JUFO Level: 2.**
- 4 **Hassan, Mir**, Yildirim, K. S., & Iacca, G. (2025b). Adaptive weighted sign-based aggregation for practical communication-accuracy trade-off in federated learning. *IEEE Transactions on Mobile Computing*. Under Review. **JUFO Level: 3.**
- 5 Zukaib, U., Cui, X., Zheng, C., **Hassan, Mir**, & Shen, Z. (2024). Meta-ids: Meta-learning based smart intrusion detection system for internet of medical things (iomt) network. *IEEE Internet of Things Journal*, 1–1. **JUFO Level: 2.**  doi:10.1109/JIOT.2024.3387294
- 6 Zukaib, U., Cui, X., **Hassan, Mir**, Harris, S., Hadi, H. J., & Zheng, C. (2023). Blockchain and machine learning in ehr security: A systematic review. *IEEE Access*, 1–1. Journal Ranking: Q1.  doi:10.1109/ACCESS.2023.3333229
- 7 Narwani, K., Lin, H., Pirbhulal, S., & **Hassan, Mir**. (2022). Towards ai-enabled approach for urdu text recognition: A legacy for urdu image apprehension. *IEEE Access*, 1–1. **JUFO Level: 1.**  doi:10.1109/ACCESS.2022.3203426
- 8 **Hassan, Mir**, Jincai, C., Iftekhar, A., & Cui, X. (2020). Future of the internet of things emerging with blockchain and smart contracts. *International Journal Advance Computer Science and Applications*, 11(6). **JUFO Level: 1.** Retrieved from  <http://dx.doi.org/10.14569/IJACSA.2020.0110676>
- 9 **Hassan, Mir**, Jincai, C., Iftekhar, A., Shehzad, A., & Cui, X. (2020). Implementation of security systems for detection and prevention of data loss/leakage at organization via traffic inspection. *arXiv preprint arXiv:2012.14111*.
- 10 Iftekhar, A., Cui, X., **Hassan, Mir**, & Afzal, W. (2020). Application of blockchain and internet of things to ensure tamper-proof data availability for food safety. *Journal of Food Quality*, 2020. **JUFO Level: 1.** Retrieved from  <https://doi.org/10.1155/2020/5385207>

Conference Proceedings

- 1 **Hassan, Mir**, Sergio, J., Giedre, S., Ali, N., Ali Hassan, S., & Kahraman, N. (2026). Zero-knowledge inspired provenance for trustless federated learning in decentralized iot systems. In *Proceedings of ieee vehicular technology conference (vtc2026-spring)*. **Accepted**, Conference Ranking: B (VTC).
- 2 Bilal, H. M., **Hassan, Mir**, & Iacca, G. (2025). Enhancing communication-efficient federated learning for human activity recognition through knowledge distillation. In *Ieee vehicular technology conference (vtc2025)* (**Accepted**). Conference Ranking: B.
- 3 **Hassan, Mir**, Raza, W., & Iacca, G. (2025). A tinymml approach for the classification of bean crop diseases. In *Ieee vehicular technology conference (vtc2025)* (**Accepted**). Conference Ranking: B.
- 4 **Hassan, Mir**, Sinan, K., & Giovanni, I. (2025). A federated multi-task learning framework with dual attention mechanisms for smart buildings. In *Proceedings of ieee vehicular technology conference (vtc2025-spring)*. **Best Paper Award**, Conference Ranking: B (VTC).
- 5 **Hassan, Mir**, Yildirim, K. S., & Iacca, G. (2025a). A genetic algorithm-based parameter selection for communication-efficient federated learning. In *Evoapplications 2025: Applications of evolutionary computation* (Vol. 15613, pp. 437–453). Conference Ranking: A.  doi:10.1007/978-3-031-90065-5_27
- 6 **Hassan, Mir**, Custode, L. L., Yildirim, K. S., & Iacca, G. (2023). Fededge: Federated learning with docker and kubernetes for scalable and efficient edge computing. In *Proceedings of the 2023 international conference on embedded wireless systems and networks* (pp. 339–344). Conference Ranking: A.
- 7 **Hassan, Mir**, Chen, J., Zhu, C., & Zukaib, U. (2022). Adoption of blockchain-based artificial intelligence in healthcare. In *2022 5th international conference on artificial intelligence and big data (icaibd)* (pp. 140–144).  doi:10.1109/ICAIBD55127.2022.9820137
- 8 Shehzad, A., Zafar, N., **Hassan, Mir**, & Shen, Z. (2019). Food item recognition and intake measurement techniques. In *Proceedings of the 2019 11th international conference on machine learning and computing* (pp. 405–411). Retrieved from  <https://doi.org/10.1145/3318299.3318379>
- 9 **Mir, Hassan**, Cui, X., Shahzad, A., & Jumani, R. (2018). Optimization of satellite link to earth station using satellite tool kit (stk). (Vol. 6, pp. 48–52).